OpenGeoSys Training 2026 – KIGAM

Lecture 0: Course Overview | OGS Introduction

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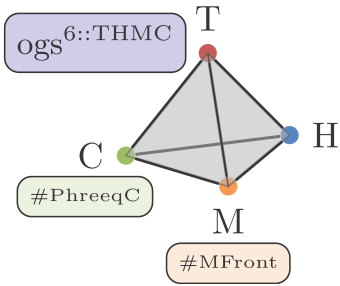
[3]Technische Universität Dresden – TUD, Dresden

Daejeon, 11th October 2026

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Agenda

1. OGS Course Overview
2. OpenGeoSys Introduction
   - OGS-THMC
   - History
   - Software Engineering
   - OGSTools
   - Benchmark Gallery
   - ModelHub
3. References

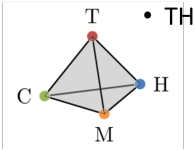


Lectures

- Continuum Mechanics
- Hydromechanics
- Geomechanics

OGS Framework

- ogs^{THMC}
- ogs^{Tools}
- ogs^{Workflows}
- ogs^{DigitalTwins}



Exercises

- ogs installation
- H: Theis: Pumping test
- H: STIMTEC: Hydraulic tests
- H: Selke: Catchment
- M: Kirsch: Excavation
- HM: Fault reactivation
- THM: FE experiment

Part 1 morning session
lectures and software demonstration

Part 2 afternoon session (optional)
hands-on work with OGS (exercises)

Olaf

Lectures

- L0 Course Overview | OpenGeoSys Introduction
- L1 Continuum Mechanics
- L2 Hydromechanics
- L3 Groundwater Modeling

Overleaf: `https://www.overleaf.com/read/gpfdtpzqrjvp#ffc4c4`

Exercises

- E0 OpenGeoSys: OGS basics, Benchmark Gallery, GitHub
- E1 Theis: Pumping well
- E2 STIMTEC: Pumping test in fractured rock
- E3 Selke: Catchment modeling

GitHub: `https://github.com/OlafKolditz/ogs-course-2026`

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Mehran

Lectures

- L4 Underground Tunnel Excavation
- L6 Hydro-Mechanically Coupled Processes in Fault-Controlled Porous Medium
- L7 Heated Tunnel Problem

Exercises

- E0 Tunnel Convergence Problem
- E1 Theis: Pumping well
- E2 STIMTEC: Pumping test in fractured rock
- E3 Selke: Catchment modeling

github:
`https://github.com/mehranqsb/OGSTrainingCourse_KIGAM_ZJU`

OpenGeoSys Introduction

OpenGeoSys THMC

Application areas



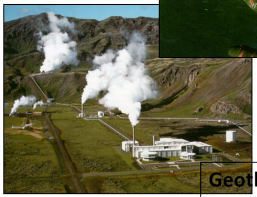
Water Resources





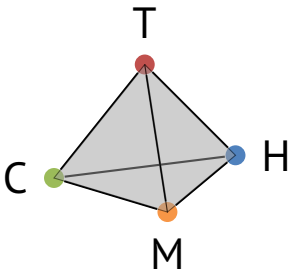
Water Quality

Geotechnical Systems





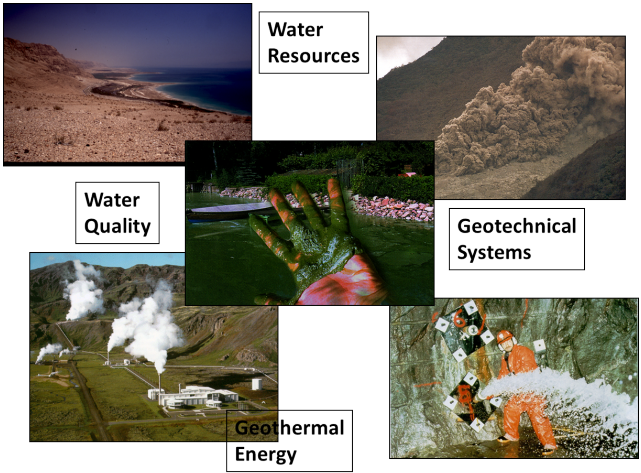
Geothermal Energy



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OpenGeoSys THMC

Framework



ogs-applications:

- T geothermal
- H geohydrological
- M geomechanical
- C geochemical (RTP)

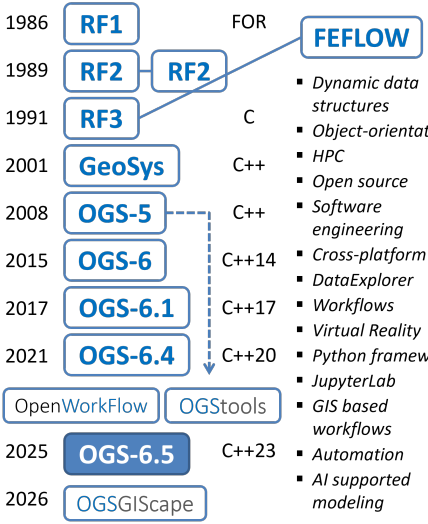
ogs-brands:

- ogs^{THMC}
- ogs^{Tools}
- ogs^{Workflows}
- ogs^{DigitalTwins}

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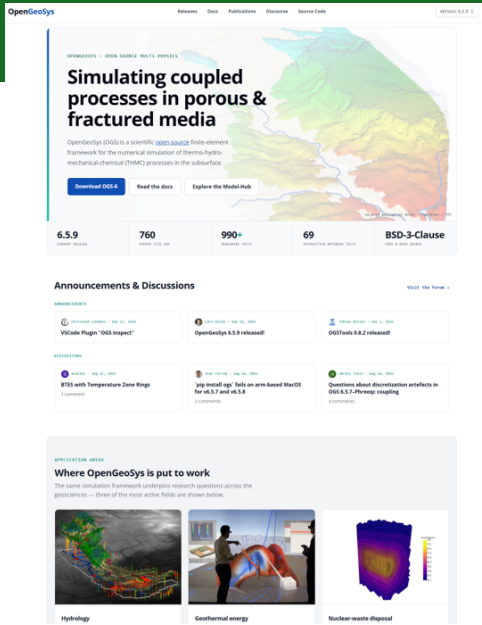
OpenGeoSys

History | Website



FEFLOW

- *Dynamic data structures*
- *Object-orientation*
- *HPC*
- *Open source*
- *Software engineering*
- *Cross-platform*
- *DataExplorer*
- *Workflows*
- *Virtual Reality*
- *Python framework*
- *JupyterLab*
- *GIS based workflows*
- *Automation*
- *AI supported modeling*



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OpenGeoSys

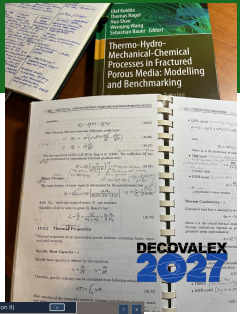
History

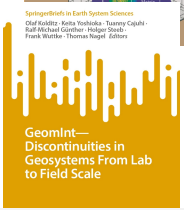
1986	RF1	FOR	FEFLOW
1989	RF2	RF2	
1991	RF3	C	
2001	GeoSys	C++	
2008	OGS-5	C++	
2015	OGS-6	C++14	
2017	OGS-6.1	C++17	
2021	OGS-6.4	C++20	
	OpenWorkFlow	OGStools	
2024	OGSGIScape	C++23	
2026	OGS-6.9		

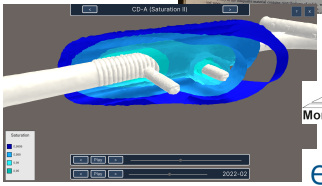
- *Dynamic data structures*
- *Object-orientation*
- *HPC*
- *Open source*
- *Software engineering*
- *Cross-platform*
- *DataExplorer*
- *Virtual Reality*
- *Python framework*





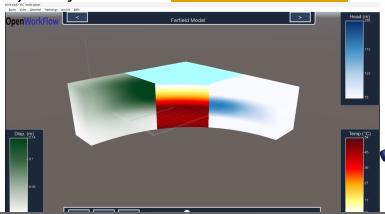


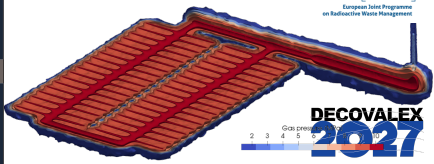


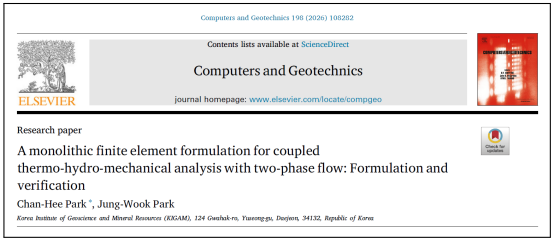












Computers and Geotechnics 190 (2026) 108282

Contents lists available at ScienceDirect

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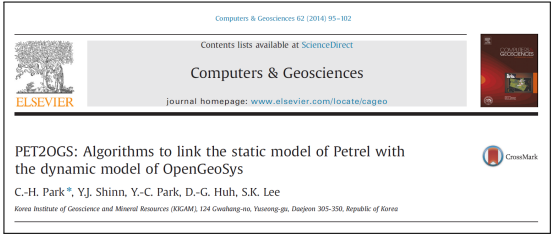
journal homepage: www.elsevier.com/locate/compgeo

Research paper

A monolithic finite element formulation for coupled thermo-hydro-mechanical analysis with two-phase flow: Formulation and verification

Chan-Hee Park *, Jung-Wook Park

Korea Institute of Geoscience and Mineral Resources (KIGAM), 124 Gwahak-ro, Yuseong-gu, Daejeon, 34132, Republic of Korea



Computers & Geosciences 62 (2014) 95–102

Contents lists available at ScienceDirect

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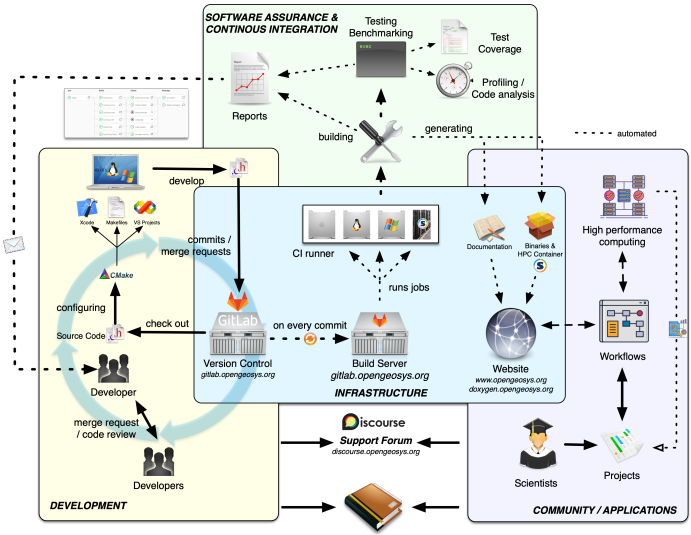
journal homepage: www.elsevier.com/locate/cageo

PET2OGS: Algorithms to link the static model of Petrel with the dynamic model of OpenGeoSys

C.-H. Park *, Y.J. Shinn, Y.-C. Park, D.-G. Huh, S.K. Lee

Korea Institute of Geoscience and Mineral Resources (KIGAM), 124 Gwahang-no, Yuseong-gu, Daejeon 305-350, Republic of Korea

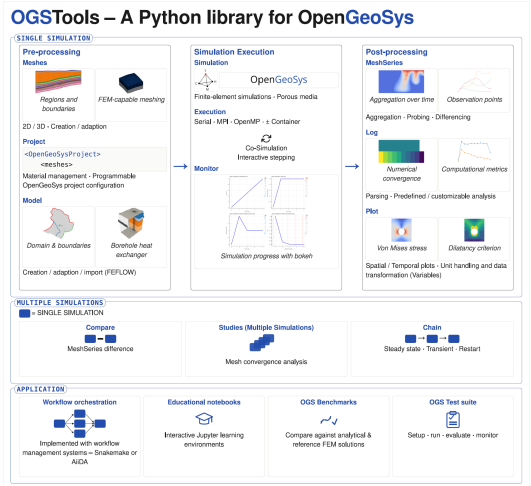
- PCH launched version in OGS 4/5 (github) and particle tracking
- OGS#PETREL interface for coupling structure and numerical models (C.-H. Park et al. 2014)
- OGS#FLAC/PFC hydro-mechanical coupling (J.-W. Park et al. 2025)
- monolithic TH2M model in OGS (Chan-Hee Park and J.-W. Park 2026)
- KIGAM is an important partner in DECOVALEX



Notes

- CI (continuous integration) system
- OGS chatbot (wip) for project files
- Guix for reproducibility
- AiiDA for provenance
- Data integration from DBs
- OGSTools for compact workflows

Ref: Bilke et al. (2025)

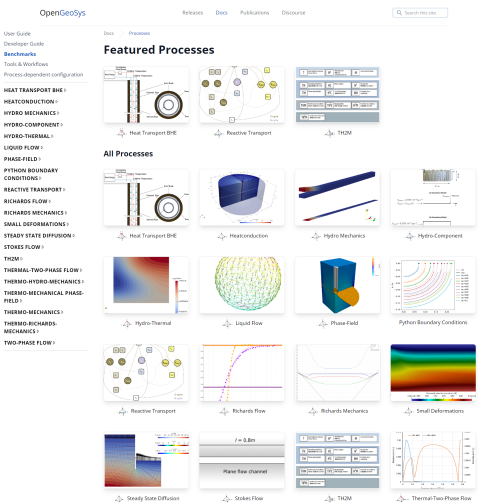


Notes

- basics for building workflows
- comprehensies can be organized by OGSTools
  - automated grid convergence
  - ensemble Monte-Carlo runs for uncertainty studies
  - LLM assistant being developed
- OGSTools is the link to AI

Ref: Meisel et al. (2026)

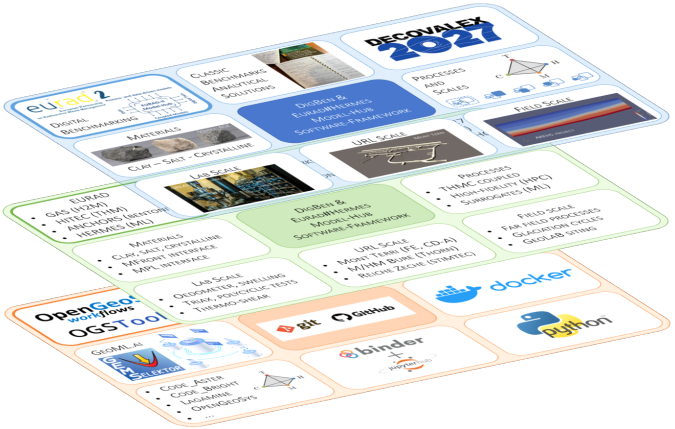
https://ogstools.opengeosys.org/stable/



Notes

- Organized by processes (THMC concept)
- Starting point for own OGS models
- Jupyter notebooks (running on ogs-binder server)
- Context and complete workflows (relying on OGSTools)

https://www.opengeosys.org/6.5.9/docs/benchmarks/



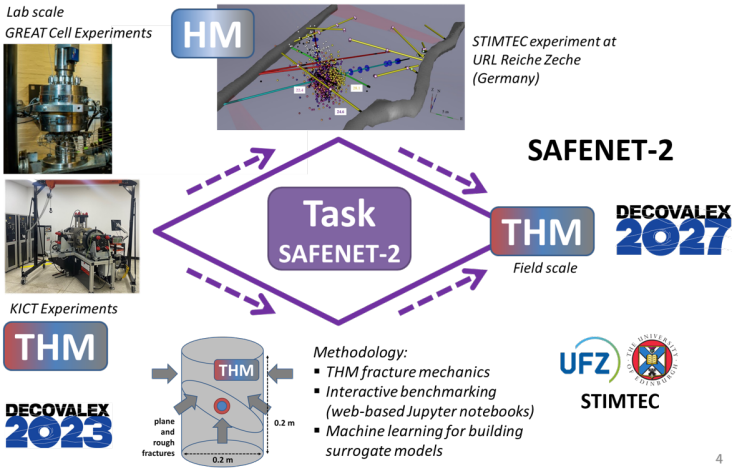
Notes

- Platform for joint benchmarking
- Initiated by EURAD and DECOVALEX projects
- Organized by application scales
- Interactive platform (via git)

Ref: Kolditz et al. (2026)

https://www.opengeosys.org/6.5.9/docs/hub/

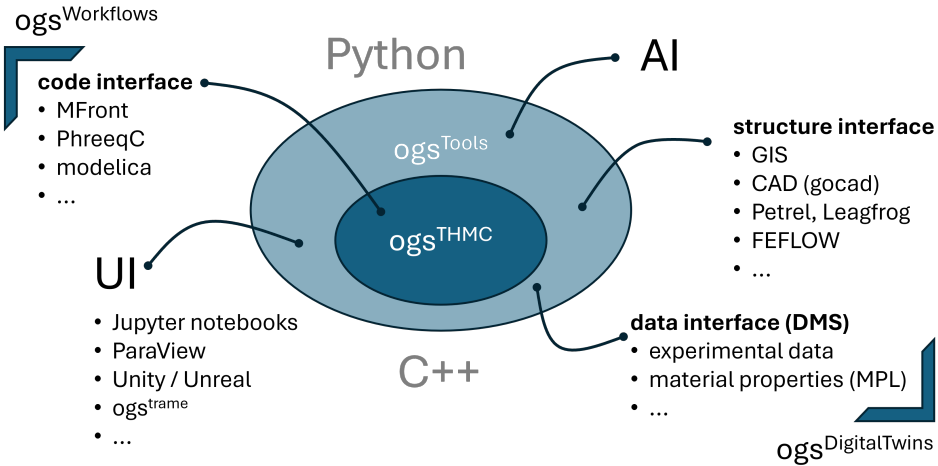
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Korean cooperation partners:

- KIGAM
- KAERI
- KICT (D2023)

STIMTEC experiment is part of the exercises


ogsWorkflows
code interface
• MFront
• PhreeqC
• modelica
• ...
Python
AI
ogsTools
ogsTHMC
structure interface
• GIS
• CAD (gocad)
• Petrel, Leagfrog
• FEFLOW
• ...
UI
• Jupyter notebooks
• ParaView
• Unity / Unreal
• ogstrame
• ...
C++
data interface (DMS)
• experimental data
• material properties (MPL)
• ...
ogsDigitalTwins

How you can get involved - become part of the community:

- providing interesting benchmarks (ideally as Jupyter notebooks) for the gallery
- trying OGS for your studies and keep record in a repo (github, gitlab etc.)
- publish your studies together with code
- participating the annual community meeting

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Exercise
OGS Installation

Exercise

OGS Installation

OGS website:
`https://www.opengeosys.org/6.5.9/`
`https://www.opengeosys.org/6.5.9/docs/userguide/basics/5-mins-ogs/`

github:
`https://github.com/mehranqsb/OGSTrainingCourse_KIGAM_ZJU`

Olaf

Lectures

- L0 Course Overview | OpenGeoSys Introduction
- L1 Continuum Mechanics
- L2 Hydromechanics
- L3 Groundwater Modeling

Overleaf: `https://www.overleaf.com/read/gpfdtpzqrjvp#ffc4c4`

Exercises

- E0 OpenGeoSys: OGS basics, Benchmark Gallery, GitHub
- E1 Theis: Pumping well
- E2 STIMTEC: Pumping test in fractured rock
- E3 Selke: Catchment modeling

GitHub: `https://github.com/OlafKolditz/ogs-course-2026`

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References

[1] Hans-Jörg G. Diersch.
FEFLOW: Finite Element Modeling of Flow, Mass and Heat Transport in Porous and Fractured Media. Springer Berlin, Heidelberg, 2014. ISBN: 978-3-642-38739-5. DOI: 10.1007/978-3-642-38739-5.

[2] Olaf Kolditz. Computational Methods in Environmental Fluid Mechanics. Springer Berlin Heidelberg, 2002. ISBN: 9783662047613. DOI: 10.1007/978-3-662-04761-3. URL: http://dx.doi.org/10.1007/978-3-662-04761-3.

[3] Olaf Kolditz, Lars Bilke, Tuanny Cajuhi, Marzieh Baes, Niels Bartels, Christoph Budach, Mauro Cacace, Holger Class, Torsten Dahm, Denise Degen, Jacob Frasunkiewicz, Jan von Harten, Thomas Heinze, Kristof Kessler, Heinz Konietzky, Yanlong Kong, Christoph Lehmann, David Lipp, Michael Dieter Martin, Tobias Meisel, Thomas Nagel, Insa Neuweiler, Jens-André Paffenholz, Felix Raith, Karsten Rink, Monika Sester, Jan Tecklenburg, Matthias Volk, Florian Wellmann, Stefan Wittek and Wenjie Xu. 'Digital twins of geosystems'. In: Environmental, Geological and Geotechnical Engineering 1.1 (2026), p. 100001. ISSN: 3117-9517. DOI: 10.1016/j.egge.2026.100001. URL: http://dx.doi.org/10.1016/j.egge.2026.100001.

[4] Olaf Kolditz, Diederik Jacques, Francis Claret, Erika Holt, Réka Szöke, David Garcia, Vanessa Montoya, Sergey V. Churakov, István Szöke, Arto Laikari, Min Chen and Tianyuan Zheng. 'Digitalisation in geosciences for environmental protection'. In: Environmental Earth Sciences 85.2 (2026). Editorial, p. 51. DOI: 10.1007/s12665-025-12750-y.

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References (cont.)

[5] Tobias Meisel, Florian Zill, Julian Heinze, Lars Bilke, Max Jäschke, Feliks Kiszkurno, Norbert Grunwald, Thomas Fischer and Jörg Buchwald. 'OGSTools: A Python package for OpenGeoSys'. In: Journal of Open Source Software 11.125 (2026), p. 10973. DOI: 10.21105/joss.10973. URL: https://doi.org/10.21105/joss.10973.

[6] C.-H. Park, Y.J. Shinn, Y.-C. Park, D.-G. Huh and S.K. Lee. 'PET2OGS: Algorithms to link the static model of Petrel with the dynamic model of OpenGeoSys'. In: Computers and Geosciences 62 (2014), pp. 95–102. DOI: 10.1016/j.cageo.2013.09.014.

[7] Chan-Hee Park and Jung-Wook Park. 'A monolithic finite element formulation for coupled thermo-hydro-mechanical analysis with two-phase flow: Formulation and verification'. In: Computers and Geotechnics 198 (2026). DOI: 10.1016/j.compgeo.2026.108282.

[8] Jung-Wook Park, Chan-Hee Park, Eui-Seob Park and Olaf Kolditz. 'A novel FEM–DEM coupling methodology for hydro-mechanical modeling in fractured geological media'. In: International Journal of Rock Mechanics and Mining Sciences 196 (2025). DOI: 10.1016/j.ijrmms.2025.106324.

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